LAB 4 – Genetic Algorithm

Delivery Route Optimization

Problem Statement

A delivery person has to visit four locations: A, B, C, and D. The person starts at location A, must visit each of the other locations exactly once, and finally return to A. The travel cost between the locations is given in the table below.

From / To	A	B	C	D
A	0	12	18	10
B	12	0	20	15
C	18	20	0	25
D	10	15	25	0

Objective

Implement Genetic Algorithm to find a route that visits every location exactly once, returns to the starting location A, and minimizes the total travel cost.

Input Format

The program should accept:

- The number of locations N.
- An N × N cost matrix.
- The starting location.

Example Input

4
0 12 18 10
12 0 20 15
18 20 0 25
10 15 25 0
Start = A

Example Output

Best Route: A → B → C → D → A
Minimum Total Cost: 67

Rules

- The route must start at A.
- Every location must be visited exactly once.
- The route must return to A.
- The objective is to minimize the total travel cost.

Requirements

1. Represent each route as a chromosome.
2. Create a small initial population of valid routes.
3. Calculate the total travel cost (fitness) of each route.
4. Select parent routes.
5. Apply crossover to create new valid routes.
6. Apply mutation by swapping two locations.
7. Repeat for a small number of generations.
8. Report the best route and its total cost.

Constraints

- Use a small problem size so that the algorithm can be demonstrated easily.
- The cost matrix contains non-negative travel costs.
- A location cannot appear more than once in the route, except for the final return to A.

Sample Test Case

Number of Locations: 4
Starting Location: A
Route to be optimized: A → ? → ? → ? → A

Sample Output

Best Route: A → B → C → D → A
Minimum Total Cost: 67

Note: The purpose of this lab is to understand the basic working of Genetic Algorithm. Keep the implementation simple and use a small number of iterations/generations.